

Can representation spreading make edits causally visible?

Token-aligned sequence-edit JEPA: curriculum VICReg screen

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The editor must care which edit we request

- ▶ The model receives a damaged worked solution and a literal insert, delete, or replace action.
- ▶ It predicts the next internal representation without reconstructing every output token.
- ▶ A collapsed model can look accurate while treating many actions as equivalent.
- ▶ Variance-Invariance-Covariance Regularization (VICReg) spreads internal states and reduces redundant features.

Question: does stronger VICReg make the transition predictor depend on the requested edit?

Five cells differ only in VICReg strength

- ▶ All cells use 2,000 curriculum-corrupted oracle trajectories, seed 0, three epochs, and zero dropout.
- ▶ All cells share the token-pointer action interface, recursive depth-four predictor, and exponential-moving-average target.
- ▶ Tested whole-loss weights are 0, .02, .1, .5, and 1.0.
- ▶ The causal test swaps actions across valid states; error must rise by at least 5%.
- ▶ Effective-rank loss must remain within 10% of the healthiest control.

Why it matters: hierarchy remains blocked until primitive edit dynamics pass both gates.

Every cell predicts, but every cell underuses the action

VICReg	One-step	No change	Shuf./match	Rank
0	.149	.256	1.013	113.6
.02	.153	.263	1.013	107.4
.1	.151	.320	1.014	84.4
.5	.149	.346	1.010	70.2
1.0	.160	.272	1.009	69.5

- ▶ Every cell beats copying the current state, so optimization is functioning.
- ▶ No shuffled-action ratio approaches the 1.05 causal threshold.
- ▶ Weight .02 is the only nonzero setting within the rank-loss gate.

The next test directly supervises action geometry

- ▶ Strong VICReg spreads values but removes 26–39% of effective rank; it is not selected.
- ▶ Weight .02 remains only as a low-cost combination anchor, not as a successful result.
- ▶ Latent Difference Action Decoding (LDAD) reconstructs the complete observed edit phrase only from the online state displacement.
- ▶ The next screen tests LDAD weights 1, 10, and 20, plus no-VICReg and moderate-VICReg interaction controls at weight 10.
- ▶ LDAD continues only if it improves causal action use and recursive prediction, not merely action-token decoding.

Limitation: this remains one-seed, candidate-privileged oracle denoising, not autonomous reasoning.